Spring 2021 Real Analysis II — Homework 3

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Denote $\mathbb{R}_+ := [0, \infty)$ and $\mathbb{R}_{++} := (0, \infty)$, $\triangle^n := \{x = (x_1, \dots, x_n) \in \mathbb{R}^n_+ : \sum_{i=1}^m x_i = 1\}$.

1. Let X be a B* space and C a proper convex subset of X such that $0 \in \text{int}(C)$. Let P be the Minkowski functional of C. Prove the following two statements: (a) $x \in \text{int}(C)$ if and only if $P(x) < 1$. (b) $\overline{\text{int}(C)} = \overline{C}$.

   Proof. (a) ($\Rightarrow$) Since $x \in \text{int}(C)$, we know there exists $\delta > 0$ such that $B(x; \delta) \subset C$, and thus

   $$\left(1 + \frac{\delta}{2\|x\|}\right) x = \frac{x}{(1 + \frac{\delta}{2\|x\|})^{-1}} \in B(x; \delta) \subset C,$$

   which implies that $P(x) \leq (1 + \frac{\delta}{2\|x\|})^{-1} < 1$.

   ($\Leftarrow$) Since P is continuous, there exists $\delta > 0$ such that $P(x') < 1$ for all $x' \in B(x; \delta)$. Hence $B(x; \delta) \subset C$ (recall that $x \in C$ if and only if $P(x) \leq 1$)) which implies that $x \in \text{int}(C)$.

   (b) It is obvious that $\overline{\text{int}(C)} \subset \overline{C}$. Note that

   $$\overline{\text{int}(C)} = \overline{\{x \in X : P(x) < 1\}} = \{x \in X : P(x) \leq 1\} \supset C,$$

   since P is continuous. Hence $\overline{C} = \overline{\text{int}(C)}$. □

2. Let X be a Banach space. If E is a finite subset of X, then $\text{conv}(E)$ is sequentially compact. [Hint: for any $\epsilon > 0$, consider a finite δ-net of $[0, 1]$ with properly chosen $\delta > 0$, and use it to construct an ϵ-net of $\text{conv}(E)$.]

   Proof. Suppose $E = \{x_1, \dots, x_m\}$. Denote $s = \sum_{i=1}^m \|x_i\| > 0$. Then for any $\epsilon > 0$, let $\Lambda^\epsilon = \{\lambda_1^\epsilon, \dots, \lambda_k^\epsilon\}$ be a finite (ϵ/s)-net of $[0, 1]$. Then for any $\lambda = (\lambda_1, \dots, \lambda_m) \in \triangle^m$, there exists $\lambda_{k_i}^\epsilon \in \Lambda^\epsilon$ such that $|\lambda_i - \lambda_{k_i}^\epsilon| < \epsilon/s$ for each $i = 1, \dots, m$. Then

   $$\left|\sum_{i=1}^m \lambda_i x_i - \sum_{i=1}^m \lambda_{k_i}^\epsilon x_i\right| \leq \sum_{i=1}^m |\lambda_i - \lambda_{k_i}^\epsilon| \|x_i\| < \epsilon.$$

   Hence $\{\sum_{i=1}^m \lambda_{k_i}^\epsilon x_i : \lambda_{k_i}^\epsilon \in \Lambda^\epsilon\}$ is a finite (with at most k^m elements) ϵ-net of $\text{conv}(E)$. Therefore $\text{conv}(E)$ is totally bounded and thus sequentially precompact. Since $\text{conv}(E)$ is closed, we know it is also sequentially compact. □

3. Let X be a Banach space. If E is a totally bounded subset of X, then $\text{conv}(E)$ is totally bounded and $\overline{\text{conv}}(E)$ is sequentially compact. [Hint: $\text{conv}(N)$ is an ϵ-net of $\text{conv}(E)$ if N is an ϵ-net of E.]

Proof. Since E is totally bounded, we know for any $\epsilon > 0$, there exists a finite ϵ-net $N^\epsilon = \{x_1^\epsilon, \dots, x_{m_\epsilon}^\epsilon\}$ of E. For any $n \in \mathbb{N}$ and $x = \sum_{i=1}^n \lambda_i x_i \in \operatorname{conv}(E)$, where $\lambda = (\lambda_1, \dots, \lambda_n) \in \triangle^n$ and $x_i \in E$, we know there exists $x_{k_i}^\epsilon \in N^\epsilon$ such that $\|x_{k_i}^\epsilon - x_i\| < \epsilon$ for each $i = 1, \dots, n$. Hence

$$\left\|\sum_{i=1}^n \lambda_i x_i - \sum_{i=1}^n \lambda_i x_{k_i}^\epsilon\right\| \le \sum_{i=1}^n \lambda_i \|x_i - x_{k_i}^\epsilon\| < \epsilon.$$

Therefore $\operatorname{conv}(N^\epsilon)$ is an ϵ-net of $\operatorname{conv}(E)$. Hence, we know for any $\epsilon > 0$, $\operatorname{conv}(N^{\epsilon/2})$ is an $(\epsilon/2)$-net of $\operatorname{conv}(E)$.

From the last problem, we know $\operatorname{conv}(N^{\epsilon/2})$ is sequentially compact. Therefore, there exists a finite $(\epsilon/2)$-net M^ϵ of $\operatorname{conv}(N^{\epsilon/2})$. Hence M^ϵ is a finite ϵ-net of $\operatorname{conv}(E)$. As ϵ is arbitrary, we know E is totally bounded. Therefore $\overline{\operatorname{conv} E}$ is also totally bounded and thus sequentially compact. □

4. Let C be a compact convex set of a B* space X and $T : C \to C$ a continuous function. Show that T has a fixed point in C.

   Proof. Since C is compact and T is continuous, we know $T(C)$ is compact and therefore sequentially compact. Now $T : C \to C$ where C is closed (since C is compact) and convex, we know by Schauder theorem there exists a fixed point of T in C. □

5. Let X be a B* space and $E \subset X$. We call $T : E \to X$ *compact* if T is continuous and $T(K)$ is sequentially compact whenever $K \subset E$ is bounded. Suppose C is a bounded closed convex set of X and $T : C \to C$ is compact. Show that T has a fixed point in C.

   Proof. Since C is bounded and T is compact, we know T is continuous and $T(C)$ is sequentially compact. Hence we know by Schauder theorem that T has a fixed point in $D \subset C$. □

6. Let $A \in \mathbb{R}_{++}^{n\times n}$. Show that there exists $\lambda > 0$ and $x \in \mathbb{R}_+^n$ such that $Ax = \lambda x$. [Hint: consider $Tx = Ax/|Ax|_1$ where $|y|_1$ is the 1-norm of the vector y.]

   Proof. Consider $Tx = Ax/|Ax|_1$, then it is easy to show that $T : \triangle^n \to \triangle^n$ is continuous. Note that $\triangle^n$ is compact and convex, we know $T(C)$ is sequentially compact and hence by Schauder theorem that T has a fixed point $x \in \triangle^n$. Therefore $Tx = x$ and thus $Ax = \lambda x$ where $\lambda = |Ax|_1$. □

7. Let $k : [0,1] \times [0,1] \to \mathbb{R}_{++}$ be continuous. Define $T : C([0,1]) \to C([0,1])$ by

   $$(Tx)(t) = \int_0^1 k(t,s)x(s)\,\mathrm{d}s$$

   for any $x \in C([0,1])$. Show that there exists $\lambda > 0$ and a nonnegative $x \in C([0,1])$ such that $Tx = \lambda x$.

   Proof. Since $k : [0,1] \times [0,1] \to \mathbb{R}_{++}$ is continuous, we know k attains minimum and maximum on $[0,1] \times [0,1]$, i.e., there exist $m, M > 0$ such that

   $$m \le k(x,y) \le M, \qquad \forall\, x,y \in [0,1].$$

   Define the set

   $$C := \left\{x \in C([0,1]) : x \ge 0,\ \int_0^1 x(t)\,\mathrm{d}t = 1\right\}.$$

Then it is easy to verify that C is closed and convex. Define $U : C \to C$ by

$$(Ux)(t) := \frac{\int_0^1 k(t,s)x(s)\,\mathrm{d}s}{\int_0^1\int_0^1 k(t,s)x(s)\,\mathrm{d}s\,\mathrm{d}t}.$$

Now we shall show that $U(C)$ is sequentially precompact. By Arzela-Ascoli theorem, it suffices to show that $U(C)$ is uniformly bounded and equicontinuous. We note that for any $x \in C$ there is

$$\|Ux\| = \frac{\|\int_0^1 k(t,s)x(s)\,\mathrm{d}s\|}{\int_0^1\int_0^1 k(t,s)x(s)\,\mathrm{d}s\,\mathrm{d}t} \le \frac{M\int_0^1 x(s)\,\mathrm{d}s}{m\int_0^1 x(s)\,\mathrm{d}s} = \frac{M}{m},$$

which implies that $U(C)$ is uniformly bounded by M/m.

Note that k is continuous on $[0,1]\times[0,1]$ and thus uniformly continuous. Hence, for any $\epsilon > 0$, there exists $\delta = \delta(\epsilon) > 0$, such that $|k(t,s) - k(t',s')| < m\epsilon$ for all $(t,s),(t',s') \in [0,1]\times[0,1]$ satisfying $|(t,s)-(t',s')| < \delta$. Therefore,

$$|(Ux)(t) - (Ux)(t')| = \frac{|\int_0^1 (k(t,s)-k(t',s))x(s)\,\mathrm{d}s|}{\int_0^1\int_0^1 k(t,s)x(s)\,\mathrm{d}s\,\mathrm{d}t} \le \frac{m\epsilon\int_0^1 x(s)\,\mathrm{d}s}{m\int_0^1 x(s)\,\mathrm{d}s} = \epsilon,$$

for all $t, t' \in [0,1]$ satisfying $|t-t'| < \delta$ and any $x \in C$. Hence C is equicontinuous.

Now we have proved that $U(C)$ is sequentially precompact. Since $U(C) \subset C$ and C is closed, we know by Schauder theorem that U has a fixed point in C, i.e., there exists $x \in C$ such that $Ux = x$, which implies that $Tx = \lambda x$ where $\lambda := \int_0^1\int_0^1 k(t,s)x(s)\,\mathrm{d}s\,\mathrm{d}t$. □

8. Let X be a Banach space and C a bounded closed convex subset of X. Suppose $T : C \to C$ satisfies

$$\|Tx - Ty\| \le \|x - y\|$$

for all $x, y \in X$. Show that $\inf_{x\in C}\|x - Tx\| = 0$. [Hint: fix $z \in C$ and $\epsilon \in (0,1)$, consider $T_\epsilon x := \epsilon z + (1-\epsilon)Tx$. Show that T_ϵ has a fixed point x_ϵ in C and $\|Tx_\epsilon - x_\epsilon\| = O(\epsilon)$.]

Proof. As C is bounded, we know there exists $M > 0$ such that $\|x\| \le M$ for all $x \in C$. Fix $z \in C$ and $\epsilon \in (0,1)$, consider $T_\epsilon x := \epsilon z + (1-\epsilon)Tx$. Then

$$\|T_\epsilon x - T_\epsilon y\| = (1-\epsilon)\|Tx - Ty\| \le (1-\epsilon)\|x-y\|, \quad \forall\, x, y \in C.$$

Hence T_ϵ is a contractive mapping on C and thus has a fixed point x_ϵ in C, i.e., $T_\epsilon x_\epsilon = x_\epsilon$. Note that

$$\inf_{x\in C}\|Tx - x\| \le \|Tx_\epsilon - x_\epsilon\| = \|Tx_\epsilon - T_\epsilon x_\epsilon\| = \|\epsilon(z - Tx_\epsilon)\| \le 2M\epsilon.$$

As $\epsilon > 0$ is arbitrary, we know $\inf_{x\in C}\|Tx - x\| = 0$. □

9. Let X be a Banach space with norm $\|\cdot\|$. Show that there exists an inner product $\langle\cdot,\cdot\rangle : X\times X \to \mathbb{R}$ such that $\langle x, x\rangle = \|x\|^2$ if and only if $\|\cdot\|$ satisfies the parallelogram law:

$$\|x+y\|^2 + \|x-y\|^2 = 2(\|x\|^2 + \|y\|^2), \quad \forall\, x, y \in X.$$

[Hint: first show that $\langle x+y, z\rangle = \langle x, z\rangle + \langle y, z\rangle$ and $\langle -x, y\rangle = \langle x, -y\rangle = -\langle x, y\rangle$ for all $x, y, z \in X$. Then show that $\langle rx, y\rangle = r\langle x, y\rangle$ for all $r \in \mathbb{R}$. Finally show that $|\langle \lambda x, y\rangle - r_k\langle x, y\rangle| \to 0$ as $r_k \to \lambda$.]

Proof. ($\Rightarrow$) Easy to verify. ($\Leftarrow$) We define for any $x, y \in X$

$$\langle x, y\rangle := \frac{1}{4}(\|x+y\|^2 - \|x-y\|^2).$$

Now we need to show this is indeed an inner product.

Step 1. We note that for any $x, y, z \in X$, there is

$$\begin{aligned}\langle x+y, z\rangle &= \frac{1}{4}(\|x+y+z\|^2 - \|x+y-z\|^2)\\ &= \frac{1}{4}\Big((2\|x+z\|^2 + 2\|y\|^2 - \|x+z-y\|^2) - (2\|x-z\|^2 + 2\|y\|^2 - \|x-z-y\|^2)\Big)\\ &= \frac{1}{4}(2\|x+z\|^2 - 2\|x-z\|^2 - \|x+z-y\|^2 + \|x-z-y\|^2).\end{aligned}$$

By exchanging x and y, we also obtain

$$\langle x+y, z\rangle = \langle y+x, z\rangle = \frac{1}{4}(2\|y+z\|^2 - 2\|y-z\|^2 + \|x+z-y\|^2 - \|x-z-y\|^2).$$

Adding the two above yields

$$\langle x+y, z\rangle = \frac{1}{4}(\|x+z\|^2 - \|x-z\|^2 + \|y+z\|^2 - \|y-z\|^2) = \langle x, z\rangle + \langle y, z\rangle.$$

Step 2. It is easy to verify that $-\langle x, y\rangle = \langle -x, y\rangle = \langle x, -y\rangle$.

Step 3. It is easy to verify that $\langle nx, y\rangle = n\langle x, y\rangle$ for any $n \in \mathbb{Z}$ by the results from the first two steps.

Step 4. For any $r \in \mathbb{Q}$, there are $m, n \in \mathbb{Z}$ and $n \neq 0$ such that $r = m/n$. Then

$$m\langle x, y\rangle = \langle mx, y\rangle = \langle nrx, y\rangle = n\langle rx, y\rangle.$$

Dividing both sides by n yields $\langle rx, y\rangle = r\langle x, y\rangle$.

Step 5. For any $\lambda \in \mathbb{R}$, there exists a sequence $\{r_k\} \subset \mathbb{Q}$ such that $r_k \to \lambda$. Hence $(\lambda - r_k)x \to 0$ for any x, and thus $\|y \pm (\lambda - r_k)x\| \to \|y\|$. Now we have

$$\Big|\langle \lambda x, y\rangle - r_k\langle x, y\rangle\Big| = \frac{1}{4}\Big|\|(\lambda - r_k)x + y\|^2 - \|(\lambda - r_k)x - y\|^2\Big| \to 0.$$

Hence $\langle r_k x, y\rangle \to \langle \lambda x, y\rangle$. On the other hand, $\langle r_k x, y\rangle = r_k\langle x, y\rangle \to \lambda\langle x, y\rangle$ since $r_k \to \lambda$. Therefore $\langle \lambda x, y\rangle = \lambda\langle x, y\rangle$. □

10. Show that there is no inner product that can induce the norm $\|x\| = \max_{0\le t\le 1} |x(t)|$ on $C([0, 1])$.

Proof. Consider $x(t) \equiv 1$ and $y(t) = t$ for $t \in [0, 1]$. Then $\|x\| = 1$, $\|y\| = 1$, $\|x+y\| = 2$ and $\|x-y\| = 1$. Hence $\|\cdot\|$ does not satisfy the parallelogram law, and thus cannot be induced by an inner product. □